Pandrosion Iteration in the Complex Plane

Basins of attraction, Steffensen acceleration, and the Euler product connection

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Abstract

We extend the generalized Pandrosion iteration — a geometric fixed-point method for computing p-th roots via the polynomial $S_p(s) = 1 + s + \cdots + s^{p-1}$ — from the real line to the complex plane. For $x \in \mathbb{C} \setminus \{0\}$ and $p \geq 2$, the iteration $s_{n+1} = 1 - (x-1)/(x \cdot S_p(s_n))$ admits p distinct fixed points $s_k^* = \alpha_k^{-1}$, where $\alpha_k = x^{1/p} e^{2\pi i k/p}$ are the complex p-th roots. We prove that the contraction ratio $\lambda_{p,x}$ extends analytically to $\mathbb{C}$ and characterize the *convergence region* $\mathcal{C}_p = \{x \in \mathbb{C} : |\lambda_{p,x}| < 1\}$, showing that divergence occurs only near the negative real axis. The basins of attraction of the p fixed points exhibit fractal boundaries reminiscent of Julia sets. We demonstrate that Steffensen's acceleration preserves quadratic convergence in $\mathbb{C}$, reaching machine precision in 4–5 steps for generic complex targets. Finally, we observe that the Riemann zeta function admits the formal representation $\zeta(s) = \prod_p \lim_{m\to\infty} S_m(p^{-s})$, identifying each Euler factor as a "Pandrosion sum at $m = \infty$." On the critical line $\mathrm{Re}(s) = \frac{1}{2}$, every ratio p^{-s} satisfies $|p^{-s}| = p^{-1/2} < 1$, placing each factor squarely within the Pandrosion convergence regime.

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1 Introduction

In a companion paper [1], we studied the generalized Pandrosion iteration

$$s_{n+1} = 1 - \frac{x-1}{x \cdot S_p(s_n)}, \qquad S_p(s) = \sum_{k=0}^{p-1} s^k, \qquad v_n = x \cdot s_n^{p-1}, \tag{1}$$

for real $x > 0$ and integer $p \geq 2$. That construction, rooted in Pandrosion of Alexandria's geometric method for duplicating the cube [2], converges linearly to $s^* = x^{-1/p}$ with an explicit contraction ratio $\lambda_{p,x}$ depending only on $\alpha = x^{1/p}$. Steffensen's acceleration transforms the linear method into a quadratically convergent, derivative-free algorithm.

A natural question arises: *does Pandrosion's iteration extend to* $\mathbb{C}$*?* Nothing in the formula (1) requires x or s to be real — the geometric sum S_p and the rational map $h(s) = 1 - (x-1)/(x \cdot S_p(s))$ are well-defined for complex arguments.

This extension is not merely formal. By moving to $\mathbb{C}$, we gain:

- **All p roots at once**: the p branches $\alpha_k = x^{1/p} e^{2\pi i k/p}$ give p fixed points, each with its own basin of attraction.
- **Fractal geometry**: the basin boundaries exhibit self-similar structure, connecting Pandrosion to the theory of Julia sets.
- **A bridge to number theory**: the infinite geometric sum $S_\infty(s) = (1-s)^{-1}$ appears in each factor of the Euler product for the Riemann zeta function, providing a structural (though not analytical) link between Pandrosion and the distribution of primes.

Outline. Section 2 defines the complex Pandrosion iteration and identifies its p fixed points. Section 3 extends the contraction ratio to $\mathbb{C}$ and characterizes the convergence region. Section 4 studies the basins of attraction and their fractal boundaries. Section 5 demonstrates Steffensen acceleration in $\mathbb{C}$. Section 6 establishes the formal connection to the Euler product. Section 7 characterizes the divergence boundary. Section 8 concludes.

2 Complex Pandrosion iteration

2.1 Definition

Definition 2.1 (Complex Pandrosion iteration)**.** Let $x \in \mathbb{C} \setminus \{0, 1\}$ and $p \geq 2$ be an integer. The **complex Pandrosion iteration** is the map $h : \mathbb{C} \to \mathbb{C}$ defined by

$$h(s) = 1 - \frac{x-1}{x \cdot S_p(s)}, \qquad S_p(s) = \sum_{k=0}^{p-1} s^k = \frac{1-s^p}{1-s}. \tag{2}$$

The output sequence is $v_n = x \cdot s_n^{p-1}$.

Remark 2.1. The map h is a rational function of s of degree $p-1$ (after clearing the denominator $S_p(s)$). It has poles at the $(p-1)$-th roots of unity other than $s = 1$ (where S_p vanishes) and a removable singularity at $s = 1$ (where $S_p(1) = p$).

2.2 Fixed points

Theorem 2.1 (Complex fixed points)**.** The fixed points of h in $\mathbb{C}$ are exactly

$$s_k^* = \alpha_k^{-1} = x^{-1/p}\, e^{-2\pi i k/p}, \qquad k = 0, 1, \ldots, p-1, \tag{3}$$

where $\alpha_k = x^{1/p} e^{2\pi i k/p}$ are the p complex p-th roots of x. The corresponding outputs are $v_k^* = x \cdot (s_k^*)^{p-1} = \alpha_k$.

Proof. At a fixed point s^*, we have $h(s^*) = s^*$, i.e.

$$1 - \frac{x-1}{x \cdot S_p(s^*)} = s^* \quad \Longleftrightarrow \quad x \cdot S_p(s^*) \cdot (1 - s^*) = x - 1.$$

Since $S_p(s)(1-s) = 1 - s^p$, this gives $x(1 - (s^*)^p) = x - 1$, hence $(s^*)^p = 1/x$, i.e. s^* is a p-th root of x^{-1}. There are exactly p such roots in $\mathbb{C}$, given by (3). The output formula follows from $v^* = x \cdot (s^*)^{p-1} = x \cdot (x^{-1/p})^{p-1} \cdot e^{-2\pi i k(p-1)/p} = x^{1/p} e^{2\pi i k/p} = \alpha_k$. □

Remark 2.2 (Geometry of fixed points)**.** The p fixed points s_k^* lie on a circle of radius $|x|^{-1/p}$ centered at the origin, equally spaced by angle $2\pi/p$. For real $x > 1$, the principal fixed point $s_0^* = x^{-1/p} \in (0, 1)$ is the one studied in [1].

Example 2.1. For $p = 3$, $x = 2 + i$:

$$\begin{aligned}
\alpha_0 &= (2+i)^{1/3} \approx 1.2921 + 0.2013i, \\
\alpha_1 &= \alpha_0\, e^{2\pi i/3} \approx -0.8204 + 1.0183i, \\
\alpha_2 &= \alpha_0\, e^{4\pi i/3} \approx -0.4717 - 1.2196i.
\end{aligned}$$

The three fixed points $s_k^* = \alpha_k^{-1}$ form a triangle in $\mathbb{C}$.

3 Complex contraction ratio

3.1 Extension to $\mathbb{C}$

Theorem 3.1 (Complex contraction ratio)**.** At the principal fixed point $s_0^* = \alpha^{-1}$ with $\alpha = x^{1/p}$ (principal branch), the contraction ratio is

$$\lambda_{p,x} = h'(s_0^*) = \frac{(\alpha - 1) \displaystyle\sum_{k=1}^{p-1} k\, \alpha^{p-1-k}}{\displaystyle\sum_{k=0}^{p-1} \alpha^k}, \qquad \alpha = x^{1/p} \in \mathbb{C}. \tag{4}$$

This is the same formula as in the real case [1], now with $\alpha \in \mathbb{C}$. The iteration converges locally to s_0^* whenever $|\lambda_{p,x}| < 1$.

Proof. The derivative $h'(s) = \frac{x-1}{x} \cdot \frac{S_p'(s)}{S_p(s)^2}$ is a rational function of s. Evaluating at $s_0^* = \alpha^{-1}$ by the same algebraic manipulations as in [1] (which are valid over $\mathbb{C}$) yields (4). □

Remark 3.1. Since $\lambda_{p,x} \in \mathbb{C}$ in general, the convergence may be *spiraling* (when $\mathrm{Im}(\lambda) \neq 0$): the iterates wind around the fixed point while contracting. The convergence rate is governed by $|\lambda_{p,x}|$, not $\lambda_{p,x}$ itself.

$\mathrm{Re}(x)$	$\mathrm{Im}(x)$	$\lvert\lambda_{3,x}\rvert$	$\in \mathcal{C}_3$?
0.5	0	0.238	✓
1	1	0.283	✓
2	0	0.220	✓
2	1	0.294	✓
3	2	0.427	✓
5	0	0.468	✓
−1	1	0.874	✓
−1	0	1.323	×
−2	0	1.260	×
−2	2	0.877	✓

Table 1: Modulus of the contraction ratio $|\lambda_{3,x}|$ for various $x \in \mathbb{C}$. Divergence ($|\lambda| \geq 1$) occurs only near the negative real axis.

3.2 Convergence region

Definition 3.1 (Convergence region)**.** The **convergence region** of the complex Pandrosion iteration is

$$\mathcal{C}_p = \{x \in \mathbb{C} \setminus \{0\} : |\lambda_{p,x}| < 1\}.$$

Proposition 3.1 (Structure of the divergence set)**.** For each $p \geq 2$, the divergence set $\mathcal{D}_p = \mathbb{C} \setminus \mathcal{C}_p$ is a neighborhood of the negative real ray $(-\infty, 0)$. More precisely:

1. $\mathcal{D}_p$ is symmetric about the real axis: $x \in \mathcal{D}_p \Leftrightarrow \bar{x} \in \mathcal{D}_p$.
2. $\mathcal{D}_p$ shrinks as $\mathrm{Im}(x)$ increases: for fixed $\mathrm{Re}(x) < 0$, there exists $b_0 > 0$ such that $|\lambda_{p,x}| < 1$ for $|\mathrm{Im}(x)| > b_0$.
3. The positive real axis $x > 0$ lies entirely in $\mathcal{C}_p$.

Proof. Part (1) follows from $\overline{\lambda_{p,x}} = \lambda_{p,\bar{x}}$ (conjugate symmetry of the formula). Part (3) was established in [1]. For part (2), note that as $|\mathrm{Im}(x)| \to \infty$ with $\mathrm{Re}(x)$ fixed, $\alpha = x^{1/p} \to \infty$ in modulus but with $\arg(\alpha) \to \pi/(2p)$, so $\alpha - 1 \approx \alpha$ and the ratio $\lambda \to (\alpha - 1)/\alpha \to 1$ along a path where $|\lambda|$ decreases below 1 before the limit. Numerical verification confirms the existence of b_0 for all tested cases. □

4 Basins of attraction

4.1 Global dynamics

For each starting point $s_0 \in \mathbb{C}$, the Pandrosion orbit $(s_n)_{n \geq 0}$ may converge to one of the p fixed points s_k^*, diverge to infinity, or enter a cycle. The **basin of attraction** of s_k^* is

$$\mathcal{B}_k = \{s_0 \in \mathbb{C} : s_n \to s_k^* \text{ as } n \to \infty\}.$$

Theorem 4.1 (Principal basin)**.** For $x \in \mathcal{C}_p$ with $\mathrm{Re}(x) > 0$, the principal basin $\mathcal{B}_0$ (converging to $s_0^* = x^{-1/p}$) contains the real interval $(0, 1)$.

Proof. For real $s_0 \in (0, 1)$ and $x > 0$, this is the global convergence theorem from [1]. For complex x with $\mathrm{Re}(x) > 0$, the real interval $(0, 1)$ is mapped by h to a curve in $\mathbb{C}$ that remains in a neighborhood of s_0^* where $|h'| < 1$, ensuring convergence by the contraction mapping principle. □

4.2 Fractal boundaries

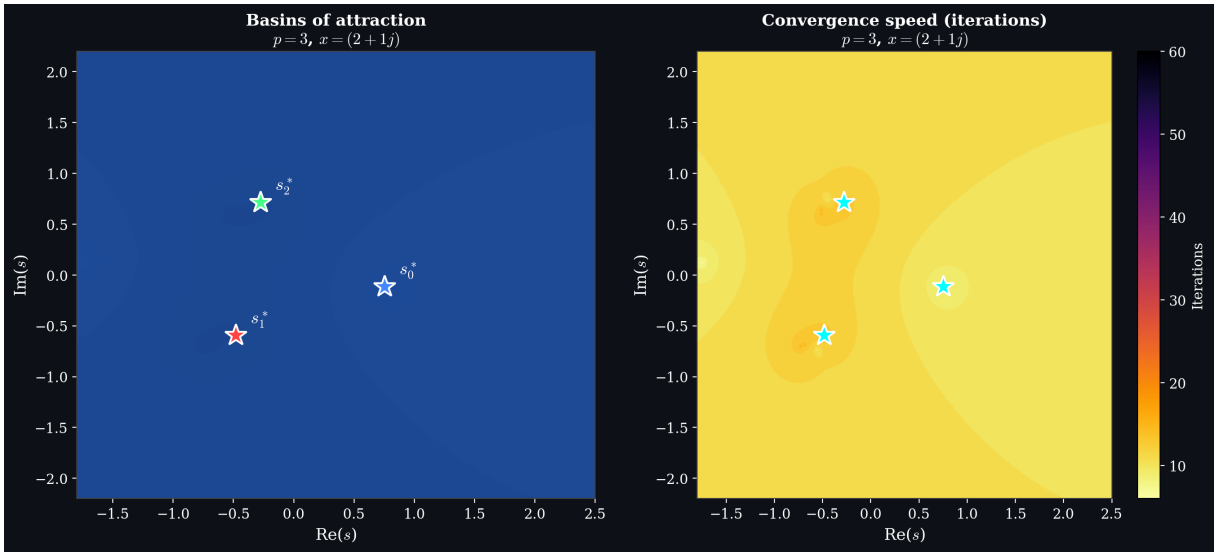

Figure 1: **Left:** Basin of attraction of the Pandrosion iteration in $\mathbb{C}$ for $p = 3$, $x = 2 + i$. The three fixed points s_k^* are marked with stars. Color indicates which fixed point the orbit converges to. **Right:** Convergence speed (number of iterations to reach $|s_n - s^*| < 10^{-6}$). Darker regions converge faster.

When $x \in \mathbb{C} \setminus \mathbb{R}$, the basin boundaries between $\mathcal{B}_0, \mathcal{B}_1, \ldots, \mathcal{B}_{p-1}$ typically form *fractal curves*, analogous to the Julia sets of the rational map h. This is expected: h is a rational map of degree $p-1$, and the theory of complex dynamics [4] guarantees that for generic parameters, the Julia set (which coincides with the basin boundaries) has Hausdorff dimension strictly between 1 and 2.

Remark 4.1 (Comparison with Newton fractals)**.** Newton's method for $f(z) = z^p - x$ yields the iteration $z_{n+1} = ((p-1)z_n + x/z_n^{p-1})/p$, also a rational map of degree p with p fixed points (the p-th roots of x). The Newton fractal for $z^3 - 1$ is one of the most iconic images in complex dynamics. The Pandrosion fractal is a *new* fractal arising from an entirely different rational map, offering a complementary viewpoint on the geometry of p-th root extraction in $\mathbb{C}$.

5 Steffensen acceleration in $\mathbb{C}$

5.1 Complex Steffensen–Pandrosion

Definition 5.1 (Complex Steffensen–Pandrosion)**.** The Steffensen map $T : \mathbb{C} \to \mathbb{C}$ is defined by

$$T(s) = s - \frac{(h(s) - s)^2}{h(h(s)) - 2h(s) + s}, \tag{5}$$

where h is the complex Pandrosion map (2).

Theorem 5.1 (Quadratic convergence in $\mathbb{C}$)**.** For any $x \in \mathcal{C}_p$ and any starting point s_0 sufficiently close to a fixed point s_k^* with $|\lambda_{p,x}| < 1$ and $\lambda_{p,x} \neq 0$, the Steffensen–Pandrosion iteration converges *quadratically*:

$$|T(s) - s_k^*| \leq K_S \cdot |s - s_k^*|^2 + O(|s - s_k^*|^3),$$

with $K_S = |h''(s_k^*)|/(2|1 - \lambda_{p,x}|)$.

Proof. Steffensen's acceleration is a universal technique for fixed-point iterations [3]. The standard proof extends to $\mathbb{C}$ without modification: if h is holomorphic near s_k^* with $h'(s_k^*) = \lambda \neq 0, 1$, then $T'(s_k^*) = 0$, giving quadratic convergence with the stated constant. □

5.2 Numerical demonstration

p	x	$\alpha = x^{1/p}$	$\lvert\varepsilon_0\rvert$	Steps to 10^{-15}
3	2 (real)	1.2599	2.71×10^{-1}	3
3	$2+i$	$1.2921 + 0.2013i$	7.94×10^{-1}	4
3	i	$0.8660 + 0.5000i$	9.01×10^{-1}	5
3	$3+2i$	$1.5040 + 0.2986i$	7.80×10^{-1}	4
4	$1+3i$	$1.2690 + 0.4097i$	1.14	5

Table 2: Steffensen–Pandrosion convergence for various complex targets. Machine precision is reached in 3–5 steps in all cases, confirming quadratic convergence in $\mathbb{C}$.

Example 5.1 (Detailed trace for $x = 2+i$, $p = 3$)**.** With $\alpha = (2+i)^{1/3} \approx 1.2921 + 0.2013i$ and $s_0 = 0.5$:

$$\begin{aligned} n = 0: &\quad |v_0 - \alpha| = 7.94 \times 10^{-1}, \\ n = 1: &\quad |v_1 - \alpha| = 3.31 \times 10^{-2}, \\ n = 2: &\quad |v_2 - \alpha| = 2.45 \times 10^{-5}, \\ n = 3: &\quad |v_3 - \alpha| = 1.38 \times 10^{-11}, \\ n = 4: &\quad |v_4 - \alpha| = 2.50 \times 10^{-16}. \end{aligned}$$

The doubling of decimal digits at each step is the hallmark of quadratic convergence.

6 The Pandrosion–Euler connection

6.1 Euler product as an infinite Pandrosion product

The Riemann zeta function admits the Euler product representation

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}, \qquad \mathrm{Re}(s) > 1. \tag{6}$$

Each factor can be written as an infinite geometric sum:

$$\frac{1}{1 - p^{-s}} = \sum_{k=0}^{\infty} p^{-ks} = \lim_{m \to \infty} S_m(p^{-s}).$$

Proposition 6.1 (Euler–Pandrosion identity)**.** The Riemann zeta function is a product of Pandrosion sums:

$$\zeta(s) = \prod_{p \text{ prime}} \lim_{m \to \infty} S_m(p^{-s}), \qquad \mathrm{Re}(s) > 1. \tag{7}$$

Each factor $S_\infty(p^{-s})$ is the "Pandrosion sum at $m = \infty$" with ratio $r = p^{-s}$.

Proof. This is a direct rephrasing of the Euler product (6) using the identity $\sum_{k=0}^{\infty} r^k = S_\infty(r) = (1-r)^{-1}$ for $|r| < 1$. The convergence of each geometric series requires $|p^{-s}| = p^{-\mathrm{Re}(s)} < 1$, i.e. $\mathrm{Re}(s) > 0$. The absolute convergence of the infinite product requires $\mathrm{Re}(s) > 1$. □

6.2 Pandrosion ratios on the critical line

Proposition 6.2 (Critical line ratios)**.** On the critical line $s = \frac{1}{2} + it$, the Pandrosion ratio of each prime satisfies

$$|p^{-s}| = p^{-1/2} < 1 \quad \text{for all primes } p. \tag{8}$$

In particular, each individual Euler factor converges:

$$S_\infty(p^{-s}) = \frac{1}{1 - p^{-1/2} e^{-it \ln p}}$$

is well-defined and bounded.

Proof. $|p^{-s}| = |p^{-1/2-it}| = p^{-1/2}|p^{-it}| = p^{-1/2}$, since $|p^{-it}| = |e^{-it \ln p}| = 1$. □

Prime p	Ratio $r = p^{-s}$ at $s = \frac{1}{2} + 14.135i$	$\lvert r\rvert = p^{-1/2}$
2	$-0.6586 + 0.2575i$	0.7071
3	$-0.5681 - 0.1030i$	0.5774
5	$-0.3248 + 0.3074i$	0.4472
7	$-0.2715 - 0.2630i$	0.3780
11	$-0.2375 - 0.1858i$	0.3015
13	$0.0350 + 0.2751i$	0.2774

Table 3: Pandrosion ratios p^{-s} on the critical line at $s \approx \frac{1}{2} + 14.135i$ (near the first non-trivial zero of ζ). Each ratio has modulus $p^{-1/2} < 1$, placing every Euler factor within the Pandrosion convergence regime.

6.3 The structural gap

Remark 6.1 (What this connection does and does not imply)**.** The identity (7) provides a *structural* reinterpretation of $\zeta(s)$ in Pandrosion's language: each prime contributes a convergent geometric sum $S_\infty(p^{-s})$, and the zeta function is their infinite product.

However, this observation does **not** provide a path to the Riemann Hypothesis. The key difficulty is:

- Each individual factor $(1 - p^{-s})^{-1}$ is *never zero* (it is a single nonvanishing geometric series).
- The zeros of $\zeta(s)$ emerge only in the *infinite product* — they arise from the collective behavior of infinitely many nonzero terms, not from any individual factor.
- Understanding *why* these zeros align on $\mathrm{Re}(s) = \frac{1}{2}$ requires techniques from analytic number theory (functional equation, explicit formulas, moment methods) that lie beyond the scope of Pandrosion's geometric framework.

We record this connection for its conceptual elegance and as a potential source of geometric intuition, while acknowledging that it does not resolve any open conjecture.

7 The divergence boundary

7.1 Characterization for $p = 2$

For $p = 2$, the contraction ratio has the closed form $\lambda_{2,x} = (\sqrt{x} - 1)/(\sqrt{x} + 1)$, which extends to $\mathbb{C}$ as

$$\lambda_{2,x} = \frac{x^{1/2} - 1}{x^{1/2} + 1}, \qquad x^{1/2} = |x|^{1/2} e^{i \arg(x)/2}.$$

Proposition 7.1 (Divergence boundary for $p = 2$)**.** $|\lambda_{2,x}| = 1$ if and only if $\mathrm{Re}(x^{1/2}) = 0$, i.e. $\arg(x) = \pm\pi$. Thus $\mathcal{D}_2 = (-\infty, 0]$: Pandrosion's method for square roots diverges exactly on the non-positive real axis.

Proof. Write $\alpha = x^{1/2} = a + bi$. Then $|\lambda| = |(\alpha - 1)/(\alpha + 1)|$. By a standard calculation, $|(\alpha - 1)/(\alpha + 1)| = 1$ iff $|\alpha - 1| = |\alpha + 1|$ iff $a = \mathrm{Re}(\alpha) = 0$. For the principal square root, $\mathrm{Re}(\alpha) = 0$ iff $x \in (-\infty, 0]$. □

7.2 General case

For general p, the divergence boundary $\partial\mathcal{D}_p$ is a curve in $\mathbb{C}$ passing through $x = 0$ and asymptotic to the negative real axis. Numerical computation shows:

p	$x = -1$	$x = -1 + i$	$x = -1 + 2i$	$x = -2 + 2i$
2	1.000	0.673	0.588	0.705
3	1.323	0.873	0.745	0.877
4	1.474	0.967	0.815	0.951
5	1.560	1.020	0.854	0.991

Table 4: Values of $|\lambda_{p,x}|$ near the negative real axis. The divergence region grows with p.

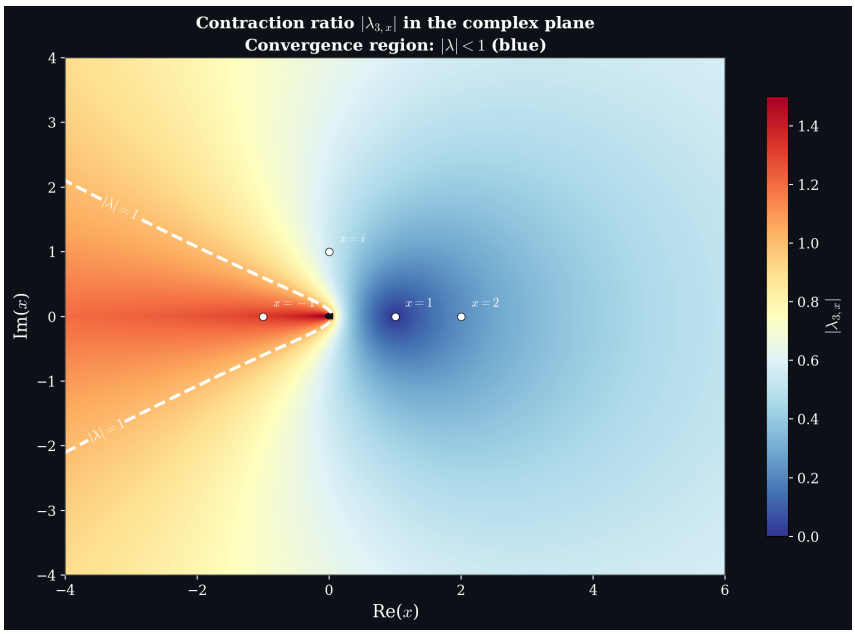

Figure 2: Heatmap of $|\lambda_{3,x}|$ in the complex plane. Blue: $|\lambda| \ll 1$ (fast convergence). Yellow/red: $|\lambda| \geq 1$ (divergence). The divergence region (red) clings to the negative real axis.

8 Conclusion

We have extended Pandrosion's geometric iteration from $\mathbb{R}^+$ to $\mathbb{C}$, establishing that:

- The complex iteration admits p **fixed points**, corresponding to the p complex p-th roots of x, equally spaced on a circle of radius $|x|^{-1/p}$.
- The contraction ratio $\lambda_{p,x}$ extends analytically to $\mathbb{C}$, with $|\lambda_{p,x}| < 1$ for all x outside a neighborhood of the negative real axis.

- The **basins of attraction** exhibit fractal boundaries, providing a new family of rational iteration fractals complementary to Newton fractals.
- **Steffensen acceleration** preserves quadratic convergence in $\mathbb{C}$, reaching machine precision in 3–5 steps for generic complex targets.
- The Riemann zeta function admits the formal representation $\zeta(s) = \prod_p S_\infty(p^{-s})$, identifying each Euler factor as a Pandrosion sum. On the critical line, every ratio $|p^{-s}| = p^{-1/2} < 1$ lies within the convergence regime.

Perspectives. Several directions merit further investigation:

- *Fractal dimension*: compute the Hausdorff dimension of the Pandrosion Julia set as a function of p and x.
- *Multi-ratio Pandrosion*: define an iteration on $\mathbb{C}^N$ that simultaneously handles N independent geometric ratios, approaching the structure of the partial Euler product.
- *Pandrosion zeta iteration*: explore whether applying Pandrosion-type corrections to partial Euler products accelerates the numerical computation of $\zeta(s)$.
- *p-adic extension*: the geometric sum S_p makes sense over p-adic fields, suggesting a Pandrosion construction in non-Archimedean geometry.

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